

## GROUP AND RING THEORY - EXERCISE SHEET 0

**0.1.** Let  $R$  be a ring. An *idempotent* is an element  $a \in R$  such that  $a^2 = a$ . Further we say that two idempotents  $e$  and  $f$  are *orthogonal* if  $ef = fe = 0$ . Let  $e$  and  $f$  be orthogonal idempotents in a ring. Show that  $e + f$  is an idempotent.

**0.2.** A ring  $R$  is (*von Neumann*) *regular* if for every  $a \in R$  there exists  $x \in R$  such that  $axa = a$ . Show that every non-zero ideal of  $R$  contains a non-zero idempotent.

**0.3.** Let  $R$  be a division ring with  $|R| = p^2$ , for some prime  $p$ . Prove that  $R$  is a field.

**0.4.** Let  $R$  be a ring with unity. Then  $R$  has only trivial one-sided ideals if and only if  $R$  is a division ring.

**0.5.** Let  $G$  be a group and suppose  $G/Z(G) \cong C_n$  for some  $n \in \mathbb{N}$ . Show that  $G$  is abelian.

**0.6.** Let  $G$  be a finite group, and let  $S, T \leq G$ . Show that

$$|ST| = \frac{|S||T|}{|S \cap T|}.$$

**0.7.** Let  $R$  be a ring.

(a) Show that the centre of  $R$ ,

$$Z(R) = \{z \in R \mid zr = rz \text{ for all } r \in R\}$$

is a subring of  $R$ .

(b) Let  $r \in R$ . Show that the centraliser

$$C_R(r) = \{a \in R \mid ar = ra\}$$

of  $r$  in  $R$  is a subring of  $R$ .

(c) Prove also that

$$Z(R) = \bigcap_{r \in R} C_R(r);$$

that is, the centre of  $R$  is the intersection of all element centralisers in  $R$ .

**0.8.** Recall that an ideal  $P \neq R$  is *prime* if whenever  $a, b \in R$  and  $ab \in P$ , then  $a \in P$  or  $b \in P$ . Show that if  $I, J$  are ideals in a commutative ring  $R$  such that  $I \not\subseteq J$  and  $J \not\subseteq I$ , then  $I \cap J$  is not prime in  $R$ .

**0.9.** Recall that an ideal  $M \subsetneq R$  is a *maximal* ideal of  $R$  if and only if  $R$  has no ideal  $I$  such that  $M \subsetneq I \subsetneq R$ . Show that  $(0)$  is a prime ideal, but not a maximal ideal, in  $\mathbb{Z}$ .

**0.10.** Recalling that  $\mathbb{Z}$  is a principal ideal domain (PID), show that  $\mathbb{Z}$  has infinitely many distinct maximal ideals.